

The elimination method



1

a 7

b $x = -3$

c $y = \frac{3}{7}$

2

a $x = -\frac{16}{3}, y = 3$

b $x = 2, y = -9$

c $x = 4, y = -9$

d $x = 4, y = -\frac{13}{3}$

e $x = -4, y = \frac{23}{3}$

f $x = 0, y = 3$

g $x = -4, y = -\frac{13}{9}$

h $x = 2, y = \frac{11}{7}$

i $x = -2, y = -2$

j $x = -20, y = \frac{53}{3}$

k $x = 3, y = -5$

l $x = -5, y = 7$

m $x = -\frac{2}{3}, y = \frac{1}{3}$

n $x = -\frac{32}{7}, y = -3$

o $x = -\frac{7}{9}, y = -2$

p $x = 1, y = -3$

q $x = 5, y = -2$

r $x = 3, y = 1$

3

a

i $3x + 11y = -48$ and $-6x + 21y = -76$

ii $x = -\frac{4}{3}$ and $y = -4$

b

i $2x + 3y = 5$ and $5x + 4y = 2$

ii $x = -2$ and $y = 3$

c

i $4x + 3y = 20$ and $8x - 3y = 4$

ii $x = 2, y = 4$

d

i $6x + 5y = 240$ and $x + 3y = 66$

ii $x = 30, y = 12$

4

a $x = 2, y = 8$

b $x = 2, y = -9$

c $x = 8, y = 3$

d $x = 5, y = 5$

e $x = -9, y = 4$

f $x = 6, y = 7$

g $x = -20, y = 15$

h $x = 3, y = \frac{23}{3}$

i $x = -1, y = -1$

j $x = \frac{1}{2}, y = \frac{3}{2}$

k $p = -\frac{3}{2}, q = \frac{23}{10}$

l $x = -2, y = 3$



Applications

5

a $x + y = 122$

b $x - y = 12$

c $x = 67$

d $y = 55$

6

a $9x + 2y = 26$

b $3x + 4y = 22$

c $x = \$2$

d $y = \$4$

7

a $x = 1$

b $y = 68$

c 2016

d 68%

8

a $2x + y = 27, 5x - y = 43$

b $x = 10, y = 7$

9

a $x + y = 10, 6x - 5y = 5$

b $x = 5, y = 5$

10

a $7x + y = 64, 3x - y = 16$

b $x = 8, y = 8$

11

a $y = 10x, y = 7x + 18$

b Son is 6 and mother is 60 years old.